

## JOROINEN VARKAUS, Finland

WMO: 027480

Lat: 62.174N

Lon: 27.860E

Elev: 87

StdP: 100.28

Time Zone: 2.00 (EUE)

Period: 99-16

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 2             | -28.8      | -25.1 | -31.8                         | 0.2 | -28.8 | -27.9 | 0.3 | -25.1 | 7.8                   | -4.4 | 7.1 | -4.4 | 0.8                   | 320  | 0.617 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.0                   | 26.2            | 18.2 | 24.2 | 17.4 | 22.9 | 16.5 | 19.6                | 23.9 | 18.5 | 22.7 | 17.5 | 21.5 | 3.2                  | 130  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.1                            | 13.1 | 21.4 | 17.0 | 12.3 | 20.3 | 15.9 | 11.4 | 19.4 | 56.3          | 23.7 | 52.6 | 22.7 | 49.5 | 21.4 | 23.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (d)                        | (e)   | (f)                | (g) | (h)                                                | (i)   | (j)        | (k)   | (l)        | (m)   | (n)        | (o)   | (p)  | (q) |
| (4) 7.3           | 6.4  | 5.8 | DB                         | -29.8 | 28.8               | 4.8 | 1.8                                                | -33.3 | 30.1       | -36.1 | 31.2       | -38.8 | 32.2       | -42.2 | 33.6 | (4) |
| (5)               |      |     | WB                         | -30.3 | 21.2               | 4.6 | 1.9                                                | -33.6 | 22.6       | -36.3 | 23.8       | -38.9 | 24.9       | -42.2 | 26.3 | (5) |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 3.8   | -8.3  | -8.8  | -4.2  | 2.7   | 9.3   | 13.7  | 16.2  | 15.1  | 10.2  | 4.1   | -0.3 | -4.7 | (6)  |
| (7)  |                                                                                  | DBStd       | 10.06 | 7.73  | 7.79  | 5.59  | 3.82  | 4.18  | 3.62  | 2.77  | 3.06  | 3.34  | 4.04  | 4.59 | 7.09 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 2814  | 567   | 527   | 441   | 219   | 64    | 7     | 0     | 3     | 39    | 184   | 308  | 455  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 5322  | 825   | 760   | 699   | 468   | 282   | 147   | 78    | 106   | 245   | 440   | 558  | 713  | (9)  |
| (10) |                                                                                  | CDD10.0     | 559   | 0     | 0     | 0     | 1     | 42    | 117   | 191   | 161   | 44    | 3     | 0    | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 25    | 0     | 0     | 0     | 0     | 1     | 7     | 11    | 6     | 0     | 0     | 0    | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 273   | 0     | 0     | 0     | 0     | 17    | 71    | 119   | 66    | 0     | 0     | 0    | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 37    | 0     | 0     | 0     | 0     | 1     | 8     | 18    | 9     | 0     | 0     | 0    | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 2.6   | 2.6   | 2.6   | 2.7   | 2.6   | 2.8   | 2.6   | 2.3   | 2.1   | 2.5   | 2.6   | 2.8  | 3.0  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 605   | 44    | 38    | 34    | 32    | 40    | 63    | 83    | 69    | 49    | 58    | 50   | 46   | (15) |
| (16) |                                                                                  | PrecMax     | 705   | 77    | 71    | 72    | 67    | 75    | 111   | 134   | 131   | 106   | 108   | 111  | 78   | (16) |
| (17) |                                                                                  | PrecMin     | 498   | 10    | 5     | 3     | 9     | 14    | 25    | 29    | 24    | 23    | 11    | 6    | 15   | (17) |
| (18) |                                                                                  | PrecStd     | 53    | 18    | 18    | 16    | 15    | 20    | 22    | 27    | 31    | 24    | 25    | 24   | 20   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 3.9   | 4.3   | 10.4  | 17.8  | 25.9  | 27.9  | 28.9  | 27.8  | 21.1  | 13.9  | 9.9  | 7.2  | (19) |
| (20) |                                                                                  |             | MCWB  | 3.2   | 2.8   | 4.9   | 11.1  | 16.1  | 18.7  | 19.6  | 20.2  | 15.5  | 11.6  | 8.7  | 6.3  | (20) |
| (21) |                                                                                  | 2%          | DB    | 2.8   | 2.8   | 6.6   | 14.3  | 21.9  | 25.6  | 26.5  | 25.1  | 19.0  | 12.1  | 8.2  | 5.5  | (21) |
| (22) |                                                                                  |             | MCWB  | 2.3   | 1.9   | 3.0   | 8.4   | 13.8  | 17.7  | 18.9  | 18.4  | 14.7  | 10.6  | 7.3  | 4.8  | (22) |
| (23) |                                                                                  | 5%          | DB    | 1.8   | 1.5   | 4.4   | 11.9  | 19.4  | 23.2  | 24.2  | 23.2  | 17.3  | 11.1  | 7.0  | 4.1  | (23) |
| (24) |                                                                                  |             | MCWB  | 1.4   | 0.9   | 2.0   | 6.5   | 12.4  | 16.5  | 17.6  | 17.0  | 14.0  | 9.8   | 6.3  | 3.4  | (24) |
| (25) |                                                                                  | 10%         | DB    | 0.8   | 0.4   | 2.9   | 9.2   | 17.2  | 21.0  | 22.8  | 21.8  | 15.9  | 10.0  | 5.7  | 2.9  | (25) |
| (26) |                                                                                  |             | MCWB  | 0.6   | 0.0   | 1.3   | 5.3   | 11.4  | 15.0  | 16.5  | 16.6  | 13.2  | 9.2   | 5.2  | 2.4  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 3.2   | 3.1   | 5.4   | 11.1  | 17.1  | 20.7  | 21.3  | 21.1  | 16.6  | 12.5  | 8.8  | 6.5  | (27) |
| (28) |                                                                                  |             | MCDB  | 3.3   | 3.9   | 8.9   | 16.5  | 23.6  | 25.2  | 25.5  | 25.9  | 19.3  | 13.6  | 9.4  | 7.0  | (28) |
| (29) |                                                                                  | 2%          | WB    | 2.0   | 1.9   | 3.4   | 9.1   | 15.4  | 18.8  | 19.6  | 19.3  | 15.5  | 11.2  | 7.3  | 4.7  | (29) |
| (30) |                                                                                  |             | MCDB  | 2.4   | 2.5   | 5.7   | 12.7  | 19.9  | 23.0  | 24.3  | 23.4  | 18.1  | 12.1  | 7.9  | 5.1  | (30) |
| (31) |                                                                                  | 5%          | WB    | 1.1   | 1.0   | 2.4   | 7.3   | 13.6  | 17.5  | 18.6  | 18.3  | 14.5  | 10.3  | 6.1  | 3.4  | (31) |
| (32) |                                                                                  |             | MCDB  | 1.3   | 1.4   | 3.9   | 10.7  | 18.1  | 22.0  | 22.8  | 22.1  | 17.0  | 11.0  | 6.6  | 3.8  | (32) |
| (33) |                                                                                  | 10%         | WB    | 0.3   | 0.0   | 1.4   | 5.7   | 11.9  | 16.0  | 17.6  | 17.2  | 13.6  | 9.1   | 5.0  | 2.2  | (33) |
| (34) |                                                                                  |             | MCDB  | 0.5   | 0.4   | 2.5   | 8.5   | 16.7  | 20.4  | 21.6  | 20.7  | 15.7  | 9.8   | 5.4  | 2.6  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR  | 6.0   | 7.2   | 8.9   | 8.9   | 10.2  | 9.4   | 10.0  | 9.4   | 8.1   | 5.2   | 4.0  | 5.0  | (35) |
| (36) |                                                                                  |             | MCDBR | 4.2   | 4.8   | 9.7   | 12.9  | 14.0  | 12.9  | 13.1  | 13.0  | 10.7  | 6.2   | 4.4  | 3.8  | (36) |
| (37) |                                                                                  |             | MCWBR | 4.2   | 4.5   | 6.8   | 7.6   | 7.2   | 6.2   | 6.4   | 6.9   | 6.9   | 4.9   | 4.1  | 3.7  | (37) |
| (38) |                                                                                  | 5% WB       | MCDBR | 4.2   | 4.5   | 8.0   | 11.5  | 11.6  | 11.0  | 11.4  | 10.9  | 9.4   | 5.1   | 4.2  | 3.7  | (38) |
| (39) |                                                                                  |             | MCWBR | 4.2   | 4.3   | 6.0   | 7.2   | 6.6   | 5.9   | 6.0   | 6.1   | 6.4   | 4.3   | 4.0  | 3.7  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.239 | 0.289 | 0.316 | 0.354 | 0.351 | 0.348 | 0.372 | 0.356 | 0.333 | 0.317 | 0.259 | N/A  | (40) |      |
| (41) |                                                                                  | taud        | 2.034 | 2.122 | 2.190 | 2.341 | 2.453 | 2.488 | 2.437 | 2.466 | 2.522 | 2.405 | 2.128 | N/A  | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 425   | 652   | 783   | 833   | 870   | 878   | 842   | 826   | 782   | 640   | 399   | N/A  | (42) |      |
| (43) |                                                                                  | Edh at Noon | 36    | 69    | 94    | 100   | 97    | 96    | 99    | 88    | 70    | 54    | 33    | N/A  | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.21  | 0.78  | 2.31  | 3.68  | 4.88  | 5.27  | 5.07  | 3.90  | 2.21  | 0.88  | 0.23  | 0.10 | (44) |      |
| (45) |                                                                                  | RadStd      | 0.03  | 0.19  | 0.30  | 0.42  | 0.55  | 0.32  | 0.49  | 0.46  | 0.25  | 0.15  | 0.04  | 0.01 | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |      | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|-------|---------|-------|------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |      | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j)  | (k)         | (l)     | (m)     | (n)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A  | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (39 neighbors) |  | +0.51 | N/A     | N/A    | N/A   | N/A     | N/A   | -175 | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page